Beyond Text-Dominance: Understanding Modality Preference of Omni-Modal Large Language Models

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Abstract

Native Omni-modal large language models (OLLMs) have shifted from pipeline architectures to unified representation spaces. However, this native integration gives rise to a critical yet underexplored phenomenon: modality preference. To bridge this gap, we first systematically quantify modality preference of OLLMs using a newly curated conflict-based benchmark and the modality selection rate metric. Our evaluation of 10 representative OLLMs reveals a notable paradigm shift: unlike the "text-dominance" of traditional VLMs, advanced OLLMs exhibit a pronounced visual preference. To further understand the underlying mechanism, we conduct layer-wise probing and demonstrate that such modality preference is not static but emerges progressively in the mid-to-late layers. Building on these insights, we conduct targeted representational interventions and reveal that modality preference causally contributes to cross-modal hallucinations, and further leverage the learned probes for hallucination detection, achieving competitive performance on downstream multimodal benchmarks without task-specific data. Our work provides both a mechanistic understanding and a practical tool for building more trustworthy OLLMs.

Code https://github.com/icip-cas/OmniPreference

1 Introduction

The recent evolution of large multimodal models has witnessed a paradigm shift from pipeline-based vision-language models (VLMs) (Bai et al., 2025; Wang et al., 2025a; Team et al., 2025) to native omni-modal large language models (OLLMs) such as Gemini-3 (Google DeepMind, 2026) and GPT-5 (Singh et al., 2025). By projecting continuous signals from diverse modalities (e.g., vision, audio and text) into a unified representation space (Team, 2026; Cui et al., 2026), OLLMs exhibit unprecedented capabilities in cross-modal reasoning (Xu et al., 2025) and seamless human-AI interaction (Tong et al., 2025). However, this native integration gives rise to a critical yet underexplored phenomenon: modality preference. When processing multimodal inputs, models often implicitly assign unequal weights to different modalities. In traditional VLMs, this is broadly perceived as a "text-dominance" bias, where models overwhelmingly rely on textual cues while overriding visual evidence (Deng et al., 2025; Zheng et al., 2025). Yet, for the emerging class of unified OLLMs, how they navigate internal modality competition remains a fundamentally unresolved black box.

Bridging this gap is crucial not only for mechanistic interpretability (Lin et al., 2025) but also for model application (Lou et al., 2025). Uncontrolled modality reliance is a primary catalyst for cross-modal hallucinations, where the model fabricates responses based on its preferred modality while ignoring factual signals from others (Leng et al., 2026, 2024; Deng et al., 2025; Bai et al., 2024). To this end, this paper conducts a systematic investigation regarding modality preference of OLLMs, aiming to answer the following 3 critical research questions (RQ):

- RQ1: How can the modality preferences of different OLLMs be systematically quantified, and what

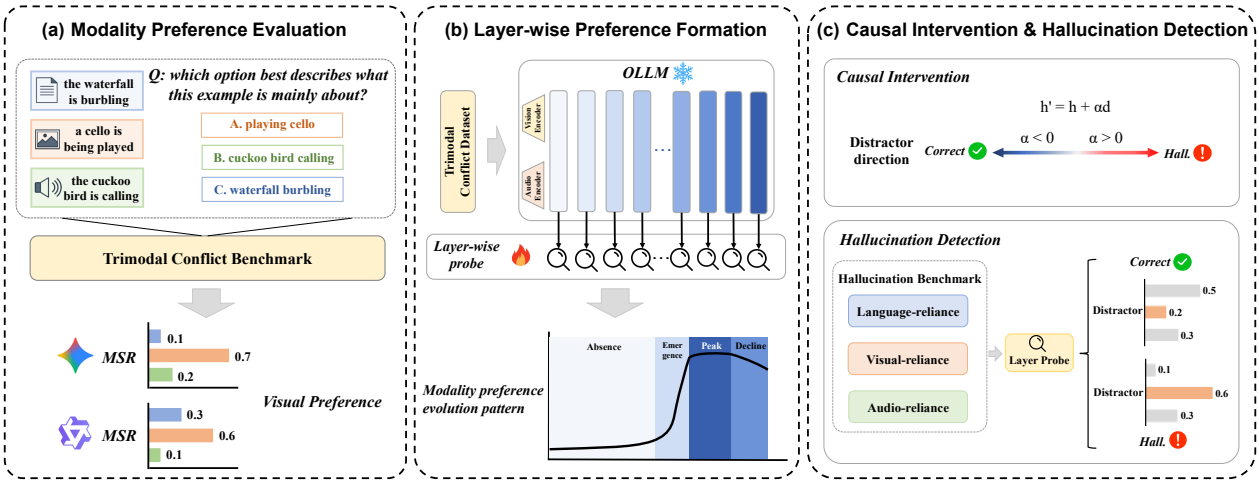


Figure 1 Overview of our analytical framework. (a) We construct a trimodal conflict benchmark to evaluate the modality preference of OLLMs. (b) We employ linear probing to characterize the layer-wise evolution of modality preference within OLLMs. (c) We further leverage the learned probes to steer the distractor modality direction, verifying that modality preference causally contributes to cross-modal hallucinations, and to detect cross-modal hallucinations.

patterns emerge across models?

- RQ2: What are the underlying mechanisms behind the formation of modality preferences in OLLMs?
- RQ3: How can insights from these mechanisms be leveraged to improve the reliability of OLLMs on downstream tasks?

For RQ1, we construct a framework to quantitatively evaluate and analyze the modality preference across different OLLMs, and surprisingly find that most advanced OLLMs exhibit a pronounced visual preference. Specifically, as illustrated in Figure 1(a), we first construct a multimodal benchmark, where each instance contains semantically conflicting information across 3 modalities: text, vision, and audio. The model is then required to select the modality it prefers. Furthermore, we introduce the Modality Selection Rate (MSR) to quantify modality preference, which is defined as the proportion of instances in which the model selects a given modality over the entire benchmark. Through extensive evaluations on 10 OLLMs across both open-source and proprietary models, we reveal a modality preference pattern that differs markedly from that of traditional VLMs: unlike the absolute text dominance observed in legacy VLMs, most OLLMs demonstrate a consistent visual preference, i.e., they tend to prefer visual information when presented with conflicting multimodal inputs. For instance, when given trimodal conflicting inputs, Gemini 3.1 Pro achieves an MSR of 72% for the visual modality, while the MSR for text is only 7%.

For RQ2, to further understand the formation of modality preference within OLLMs, we conduct layer-wise probing and reveal the evolutionary dynamics of modality preference. Specifically, as illustrated in Figure 1(b), we construct a multimodal training dataset with semantically conflicting inputs across 3 modalities. For each layer, we extract hidden states and train a single-layer MLP as a linear probe to predict the model's final modality preference. The probe accuracy on a held-out test set is then used to quantify the extent to which each layer encodes modality preference. Our analysis reveals a clear emergent pattern: modality preference is not static, but instead arises abruptly and stabilizes in the mid-to-late layers as representations become increasingly abstract.

To this end, for RQ3, we conduct an in-depth investigation into the relationship between modality preference and hallucinations, and find that modality preference causally contributes to cross-modal hallucinations. We further demonstrate that the learned modality preference probes serve as practical and valuable tools for detecting cross-modal hallucinations, without requiring any task-specific downstream data. Specifically, we conduct experiments on 2 widely used hallucination benchmarks: CMM and AVHBench, covering multiple cross-modal hallucination settings such as language reliance, visual reliance, and audio reliance. We first

observe that, the occurrence of hallucinations is accompanied by an abnormal increase in the predicted preference probability for the distractor modality. We then intervene along the distractor modality preference direction, and find that suppressing it reduces cross-modal hallucinations whereas amplifying it has the opposite effect, demonstrating that modality preference causally contributes to cross-modal hallucinations. Furthermore, we leverage the probe-estimated distractor modality preference probability for hallucination detection and find that it serves as an effective signal for detecting hallucinations. For instance, on the Audio Dominance task in CMM, the average AUROC score for hallucination detection based on our probe across the 3 OLLMs reached 83%.

In summary, our major contributions are summarized as follows:

- We establish a systematic framework dedicated for quantifying the modality preferences of OLLMs, introducing a newly curated benchmark alongside specialized metrics.
- Through comprehensive evaluations, we reveal that OLLMs exhibit an inherent preference for visual input.
- We trace the origins of modality preferences to the representational level and characterize their emergence patterns across the model's internal architecture.
- Leveraging these insights, we demonstrate that modality preference causally contributes to cross-modal hallucinations and that our layer-wise probes effectively detect cross-modal hallucinations in downstream applications.

This work not only provides a novel perspective for deciphering the internal decision-making mechanisms of OLLMs but also establishes an empirical foundation for building more trustworthy and hallucination-resistant artificial intelligence systems.

2 Related Work

2.1 Modality Preference

Understanding modality preference in multimodal large language models is essential for building reliable multimodal systems, and increasing research efforts have been devoted to this topic. Some studies construct conflicting image-text benchmarks to probe which modality the model favors under disagreement (Hua et al., 2025; Pezeshkpour et al., 2025; Deng et al., 2025), while others employ causal mediation analysis (Chen et al., 2024b) or gradient-based diagnostics (Kwon et al., 2025) to trace the bias within model internals. Representation engineering has also been used to evaluate modality preference across VLMs (Zhang et al., 2026). Despite diverse methodologies, these studies converge on a consistent finding: VLMs exhibit a pronounced tendency to over-rely on the text modality. However, existing modality preference research remains predominantly confined to the vision-language setting. Consequently, these conclusions may not generalize to OLLMs that integrate a broader range of modalities. Our work systematically investigates modality preference in OLLMs, filling this research gap.

2.2 Model Probing

Model probing has established itself as a reliable paradigm for interpreting learned representations in neural networks. Linear probing classifiers applied to frozen model activations have been widely adopted to decode syntactic and semantic properties in pretrained language models (Tenney et al., 2019; Belinkov, 2022). Probing has also been extended to examine factual knowledge storage across transformer layers (Petroni et al., 2019; Meng et al., 2022). Beyond language, analogous techniques have been applied to vision encoders (Alain and Bengio, 2016) and multimodal vision-language representations (Parcalabescu et al., 2022). Recent work further demonstrates that probing reliably recovers cross-modal alignment signals in joint embedding spaces (Zhang et al., 2025b). Building on these established foundations, we extend layer-wise probing to OLLMs to examine how modality preference emerges internally within OLLMs.

3 Framework Design

In this section, we propose a framework for evaluating the modality preferences of OLLMs, including a formal definition of modality preference, the construction of trimodal conflict data, and quantitative evaluation metrics.

3.1 Problem Formulation

Inspired by prior work on modality preference in bimodal settings (Deng et al., 2025; Zhang et al., 2025c), we extend this line of inquiry to the trimodal setting. Specifically, we introduce a trimodal conflict setting in which text, vision, and audio convey mutually conflicting semantic information. Under such conflicts, the model tends to generate outputs aligned with the information conveyed by its preferred modality. Formally, given an OLLM $\mathcal{O}$ and a query q comprising inputs from 3 conflicting modalities $(m_{\mathrm{t}}, m_{\mathrm{v}}, m_{\mathrm{a}})$, the model's output $\hat{y} = \mathcal{O}(q)$ is expected to align with one of the three modalities. Accordingly, we define the preference of model $\mathcal{O}$ for modality m_i as the following conditional probability:

$$P\big(\hat{y} \sim m_i \mid \mathrm{conflict}(m_{\mathrm{t}}, m_{\mathrm{v}}, m_{\mathrm{a}})\big) \tag{1}$$

3.2 Data Construction

Conflict Construction. We construct our trimodal conflict benchmark (Figure 1(a)) based on the XMod-Bench (Wang et al., 2025b), which provides semantically aligned triplets across three modalities (text, vision, and audio) with a shared ground-truth label. To introduce controllable conflicts, we first categorize all samples based on their semantic labels, consolidating them into 6 major categories: Animals, Human Activities, Music, Appliances & Machinery, Vehicles & Traffic, and Natural Sounds. We define a conflict triplet as a sample (x_i^T, x_j^V, x_k^A) in which each modality is sourced from a distinct category, i.e., $c_i \neq c_j \neq c_k$. We enumerate all $\binom{6}{3} = 20$ valid category triplets (c_i, c_j, c_k) and apply balanced sampling within each triplet, drawing an equal number of conflict samples per category combination.

Sample Formulation. We employ a fixed set of templates to convert each text label into a natural-language statement (e.g., "*bird squawking*" $\rightarrow$ "*the bird is squawking*"), which serves as the text input. Each constructed sample is then paired with a standardized question: *"Which option best describes what this example is mainly about?"* This prompt is intentionally modality-agnostic, neither privileging nor suppressing any particular modality. The 3 candidate options are the semantic labels of c_i, c_j, and c_k, presented in randomized order. Since each option is grounded in exactly one modality, the model's selection directly reveals its modality preference.

3.3 Quantitative Metrics

We introduce the Modality Selection Rate (MSR) as a metric for quantifying modality preference. Let $\mathcal{M}$ denote the set of modalities. In the trimodal conflict input setting, $\mathcal{M} = \{t, v, a\}$, corresponding to text, vision, and audio, respectively. For a modality $m \in \mathcal{M}$, $\mathrm{MSR}(m)$ is defined as the proportion of conflict samples for which the model's response aligns with m:

$$\mathrm{MSR}(m) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big[\hat{y}_i = \mathrm{opt}_i(m)\big] \tag{2}$$

where N is the total number of conflict samples, $\mathrm{opt}(m)$ denotes the candidate option assigned to modality m, $\hat{y}_i$ is the model's selection on the i-th sample, and $\mathbf{1}[\cdot]$ is the indicator function. With $|\mathcal{M}|$ modalities, a uniform baseline corresponds to $\mathrm{MSR}(m) = 1/|\mathcal{M}|$ for all m. $\mathrm{MSR}(m) > 1/|\mathcal{M}|$ indicates a model preference for modality m.

4 Landscape of Modality Preference

In this section, we conduct a comprehensive evaluation of the modality preferences exhibited by OLLMs based on our proposed framework.

4.1 Experimental Setup

OLLMs. We evaluate 10 OLLMs, including open-source models Qwen3-Omni-30B-A3B-Instruct, Qwen2.5-Omni-3B, Qwen2.5-Omni-7B (Xu et al., 2025), Ming-Lite-Omni 1.5 (AI et al., 2025), MiniCPM-o 2.6 (Yao et al., 2024), and OmniVinci (Ye et al., 2025), as well as closed-source models, Gemini 3.1 Pro, Gemini 3 Flash, Gemini 2.5 Pro, and Gemini 2.5 Flash (Comanici et al., 2025).

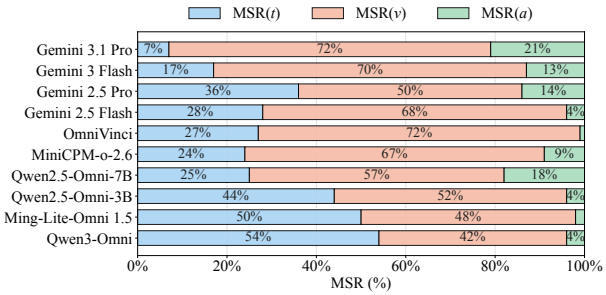


Figure 2 MSR (%) results of all evaluated OLLMs on the trimodal conflict benchmark. Qwen3-Omni refers to Qwen3-Omni-30B-A3B-Instruct.

Implementation Details. For open-source models, we perform inference with temperature $T = 0$ to obtain deterministic outputs. For closed-source models, due to network accessibility constraints, we access them through the OpenRouter (OpenRouter, Inc., 2026) API, which routes requests to the official model endpoints. We strictly follow the input formats specified in the official documentation of each model, and keep all generation parameters at their default configurations to ensure reproducibility. Following the Section 3.2, we construct a trimodal conflict benchmark. To eliminate the influence of differences in modality understanding capability, we retain only samples that the model answered correctly when each modality was presented independently, ensuring that it could correctly interpret every input modality. Taking the intersection of the samples retained by all models, we obtain a shared set of 1,000 samples for the modality preference evaluation.

4.2 Overall Results

We evaluate each model using the trimodal conflict benchmark and compute its MSR for each modality. The results are shown in Figure 2. The conclusions are as follows.

Most OLLMs exhibit a clear visual preference. Among all evaluated OLLMs, Ming-Lite-Omni 1.5 and Qwen3-Omni-30B-A3B-Instruct have slightly higher MSR(t) than MSR(v), indicating a slight preference for text. Notably, the remaining 8 models all achieve an MSR(v) exceeding 50%, with Gemini 3.1 Pro reaching as high as 72%. This suggests that, unlike the text-dominant modality preference observed in traditional VLMs (Deng et al., 2025), the majority of OLLMs exhibit a pronounced visual preference, implying that they tend to place greater trust in visual information when faced with trimodal inputs.

OLLMs exhibit insufficient utilization of audio. Across all evaluated OLLMs, MSR(a) does not exceed 21% and remains below 15% for most models; notably, Ming-Lite-Omni 1.5 and OmniVinci achieve an MSR(a) of only 2% and 1%, respectively. These results indicate that OLLMs generally rely less on the audio modality when processing multimodal inputs, suggesting insufficient utilization of audio information. Meanwhile, Gemini 3.1 Pro achieves the highest MSR(a) of 21%, indicating that more advanced models possess a greater capacity to utilize audio information.

5 How Modality Preference Emerges

In this section, we first introduce the training process of the layer-wise modality preference probes. We then reveal the internal evolution of modality preferences within OLLMs.

5.1 Layer-wise Probing Methodology

We train a single-layer MLP as a linear probe on each decoder layer of the OLLM to quantify how modality preference evolves across network depth. The overall training pipeline is illustrated in Figure 3. Consider an OLLM with L transformer decoder layers. For the i-th input sample, let $\mathbf{h}_i^{(\ell)} \in \mathbb{R}^d$ denote the hidden state at layer $\ell \in \{1, \ldots, L\}$, where d is the hidden dimension of the model.

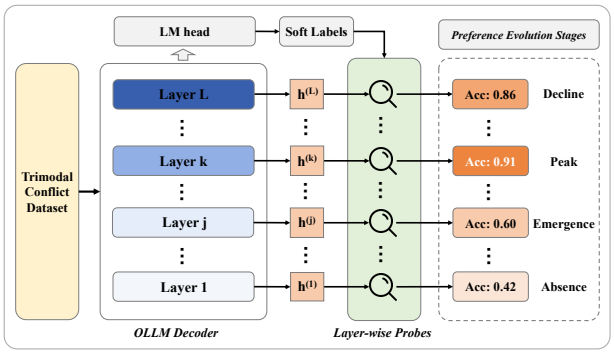


Figure 3 Illustration of the training pipeline for layer-wise modality preference probes.

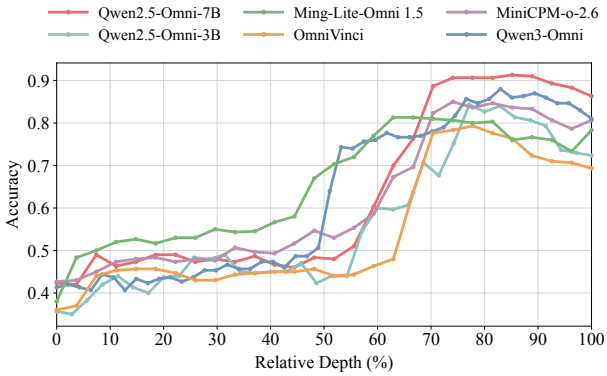


Figure 4 Probe accuracy across relative depth. Qwen3-Omni refers to Qwen3-Omni-30B-A3B-Instruct.

The last token position aggregates contextual information from the input sequence (Lin et al., 2026; Chen et al., 2024a); we therefore extract the hidden state at the last token position as the layer-wise representation for each sample. Prior to probe training, each hidden state is L_2-normalized:

$$\tilde{\mathbf{h}}_i^{(\ell)} = \frac{\mathbf{h}_i^{(\ell)}}{\left\|\mathbf{h}_i^{(\ell)}\right\|_2} \tag{3}$$

which removes magnitude variation across layers, ensuring probe captures the directional structure of the representations rather than their scale. Compared with hard labels, soft labels provide richer supervisory signals (Andonian et al., 2022). We extract the probabilities assigned to the 3 option tokens corresponding to each modality from the full-vocabulary softmax distribution at the final prompt token position, and compose them into a 3-dimensional vector serving as the soft label $\mathbf{y}_i \in \mathbb{R}^C$ $(C = 3)$ for each sample.

At each layer ℓ, the probe maps the normalized representation $\tilde{\mathbf{h}}_i^{(\ell)}$ to a predicted distribution $\hat{\mathbf{y}}_i^{(\ell)} \in \mathbb{R}^C$ over modality categories via

$$\hat{\mathbf{y}}_i^{(\ell)} = \mathrm{softmax}\big(\boldsymbol{\theta}^{(\ell)\top}\tilde{\mathbf{h}}_i^{(\ell)} + \boldsymbol{b}^{(\ell)}\big) \tag{4}$$

where $\boldsymbol{\theta}^{(\ell)} \in \mathbb{R}^{d\times C}$ and $\boldsymbol{b}^{(\ell)} \in \mathbb{R}^C$ are the learnable parameters of the probe at layer ℓ. The probe is optimized by minimizing the soft cross-entropy loss over N training samples:

$$\mathcal{J}\big(\boldsymbol{\theta}^{(\ell)}, \boldsymbol{b}^{(\ell)}\big) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\,\log \hat{y}_{i,c}^{(\ell)} \tag{5}$$

where $y_{i,c}$ and $\hat{y}_{i,c}^{(\ell)}$ denote the c-th element of the soft label $\mathbf{y}_i$ and the predicted distribution $\hat{\mathbf{y}}_i^{(\ell)}$, respectively.

To train and evaluate the linear probes, we construct a trimodal conflict dataset following the procedure introduced in Section 3.2. For each model, we independently sample 1,000 instances per modality category, yielding a class-balanced pool of 3,000 samples in total. The pool is partitioned into training, validation, and test sets at an 8:1:1 ratio, with class balance strictly maintained within each split.

5.2 How Preference Emerges

We train layer-wise modality preference probes on 6 open-source OLLMs and plot the probe accuracy as a function of relative depth. The results are presented in Figure 4. The conclusions are as follows:

Modality preference evolves through 4 stages, emerging in the mid-to-late layers and subsequently stabilizing. Modality preference evolves through 4 stages: Absence, Emergence, Peak, and Decline.

- Absence: In the first 40% of layers, probe accuracy for all models remains at a relatively low level, approximately 0.30 to 0.56, indicating that these layers primarily encode low-level features and have not yet developed clear modality preference signals.

- Emergence: Between 40% and 70% depth, all models undergo a sharp increase in probe accuracy, with modality preference signals rising rapidly within this interval. For instance, Qwen2.5-Omni-7B jumps from 0.48 to 0.89, and MiniCPM-o-2.6 rises from 0.49 to 0.82, while Qwen3-Omni-30B-A3B-Instruct completes a jump from 0.47 to 0.78 within the same interval.
- Peak: Between 70% and 90% depth, probe accuracy stabilizes with minor fluctuations across all OLLMs, where Qwen2.5-Omni-7B attains the highest accuracy of 0.91, followed by Qwen3-Omni-30B-A3B-Instruct with a peak of 0.88. This indicates that modality preference signals are the most pronounced across all layers and remain relatively stable at this stage.
- Decline: Beyond 90% depth, probe accuracy begins to decline to varying degrees across all OLLMs. This decline in the final layers aligns with prior findings that the last layers tend to compress intermediate representations into task-specific output distributions, attenuating modality-specific signals that peak at earlier depths (Tenney et al., 2019; Skean et al., 2025).

Larger OLLMs develop modality preference signals at earlier layers, smaller OLLMs exhibit more pronounced preference decay. Among all evaluated OLLMs, Qwen3-Omni-30B-A3B-Instruct and Ming-Lite-Omni 1.5 enter the Emergence stage the earliest: the former starts at 50% relative depth, while the latter enters this stage even earlier, at 40% relative depth, with its probe accuracy reaching 0.55 during the Absence stage. In contrast, OmniVinci does not enter the Emergence stage until 63% relative depth. This suggests that larger models tend to develop modality preference signals at shallower relative depths. On the other hand, Qwen2.5-Omni-3B, the smallest model evaluated, exhibits the largest accuracy decline from a peak of 0.84 to 0.72 during the Decline stage, a drop of 0.12, whereas its larger counterpart, Qwen2.5-Omni-7B, shows a decrease of only 0.05, while Ming-Lite-Omni 1.5 exhibits a decline of merely 0.03. This indicates that smaller models tend to exhibit more pronounced preference decline during the Decline stage.

6 Modality Preference and Cross-modal Hallucination

This section first analyzes the correlation between modality preference and cross-modal hallucinations. We then validate the causal effect of modality preference on hallucination behavior through representational interventions. Finally, we demonstrate that modality preference probes can serve as effective tools for hallucination detection.

6.1 Experimental Setup

The preceding analysis shows that linear probes can capture modality preference signals from the internal representations of OLLMs. A natural follow-up question is whether these internal preferences affect model reliability on downstream tasks. Prior studies identify over-reliance on a modality as a major catalyst for hallucinations (Leng et al., 2026, 2024). Based on this observation, we hypothesize that the occurrence of hallucinations is accompanied by abnormal shifts in the model's internal modality preference signals. To test this hypothesis, we conduct experiments on 2 widely used hallucination benchmarks: CMM (Leng et al., 2026) and AVHBench (Sung-Bin et al., 2024). For CMM, we adopt its Over-reliance on Unimodal Priors subcategory (Language Dominance, Visual Dominance, and Audio Dominance). For AVHBench, we adopt its 2 tasks, Video-driven Audio Hallucination and Audio-driven Video Hallucination. To establish a unified analytical framework, we define a modality role for each task: the distractor modality. Distractor modality refers to the modality that may induce hallucinations when the model over-relies on it. The role definitions for all tasks are provided in Table 1.

Task	Distractor
CMM(Language Dominance)	Text
CMM(Visual Dominance)	Vision
CMM(Audio Dominance)	Audio
AVHBench(Video-driven Audio Hall.)	Vision
AVHBench(Audio-driven Video Hall.)	Audio

Table 1 Definitions of the distractor modalities.

6.2 Modality Preference Correlates with Hallucination

We use the probe from the layer exhibiting the strongest modality preference signal in Qwen2.5-Omni-7B

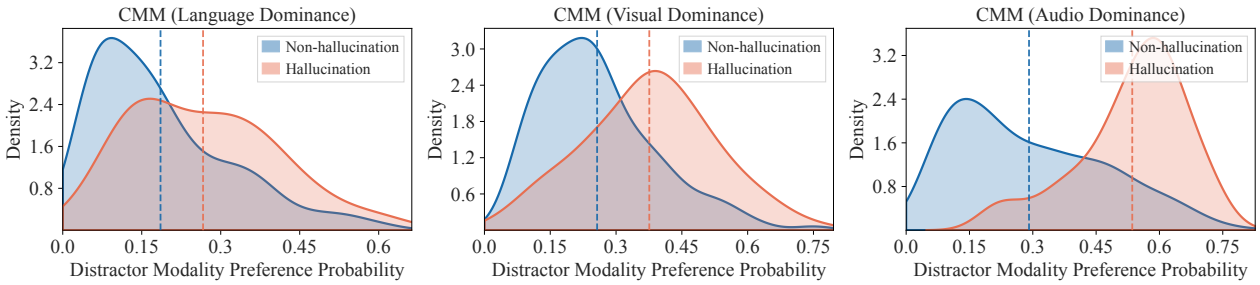


Figure 5 Probability density distributions of the distractor modality preference probabilities for non-hallucination and hallucination samples. The dashed vertical lines indicate the mean preference probabilities of the groups.

to compute the predicted preference probability assigned to the distractor modality for samples from the 3 tasks: Language Dominance, Visual Dominance, and Audio Dominance. We then plot the probability density distributions separately for non-hallucination and hallucination samples. The results are shown in Figure 5. To quantify this relationship, we conduct the Mann–Whitney U test to assess the distributional difference in distractor modality preference probability between non-hallucination and hallucination samples. The results are shown in Table 2.

Task	p-value
CMM(Language Dominance)	1.82×10^{-6}
CMM(Visual Dominance)	3.99×10^{-12}
CMM(Audio Dominance)	1.05×10^{-30}

Table 2 Mann–Whitney U test results.

Hallucination is accompanied by an abnormal increase in the distractor modality preference probability. Across all 3 tasks (Language Dominance, Visual Dominance, and Audio Dominance), the distractor modality preference probabilities of non-hallucination samples are concentrated in the low-probability region, whereas those of hallucination samples are systematically shifted toward higher values. Moreover, the mean distractor modality preference probability is consistently higher for hallucination samples than for non-hallucination samples. These results suggest that hallucinations are more likely to occur when the model exhibits an abnormal internal preference for the distractor modality. This distributional difference is also statistically significant: the Mann–Whitney U tests yield p-values well below 0.05 for all 3 tasks, indicating a strong association between an abnormal increase in distractor modality preference probability and hallucination occurrence.

6.3 Modality Preference Causally Contributes to Hallucination

Inspired by recent work on representation editing and activation steering (Deng et al., 2026; Zhang et al., 2025a; Xin et al., 2025), we leverage the trained linear probes to construct steering directions for modality preference, and perform interventions on the model's internal representations at inference time. We further investigate whether modality preference causally contributes to cross-modal hallucinations. Specifically, we select the layer ℓ^* exhibiting the strongest modality preference signal as the intervention layer. In the probe parameter matrix $\boldsymbol{\theta}^{(\ell^*)} \in \mathbb{R}^{d \times C}$ of this layer, each column vector $\boldsymbol{\theta}_m^{(\ell^*)}$ characterizes the linear direction in the representation space along which the model prefers modality m, and thus serves as the steering direction for that modality. For each task, following the definitions in Table 1, we extract the distractor modality direction $\boldsymbol{d}$, which is L_2-normalized. At inference time, we modify the hidden state $\boldsymbol{h}$ at the last token position of layer ℓ^* as:

$$\boldsymbol{h}' = \boldsymbol{h} + \alpha \boldsymbol{d} \tag{6}$$

where α is a scalar coefficient that controls the direction and strength of the intervention: $\alpha > 0$ enhances the distractor modality direction, whereas $\alpha < 0$ suppresses it. We evaluate 7 steering coefficients within $[-0.7, 0.7]$ and report the task accuracy for each setting. Figure 6 presents the results of Qwen2.5-Omni-7B on 2 tasks.

Interventions along the distractor-modality preference direction lead to changes in hallucination behavior. On Video-driven Audio Hallucination, accuracy increases monotonically as α becomes negative. At

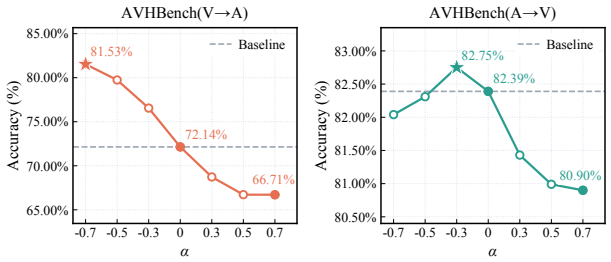


Figure 6 Task accuracy of Qwen2.5-Omni-7B under 7 steering coefficients α ranging from -0.7 to 0.7, the dashed line indicating the Baseline (α=0).

Steering Direction	α	Accuracy
Baseline	0	49.47%
Visual	−0.7	56.14%
Audio	−0.3	49.21%

Table 3 OmniBench accuracy using the best performing steering coefficient on hallucination tasks.

Task	Model	AUROC
CMM(Lan Dom)	Qwen2.5-Omni-7B	0.68
	MiniCPM-o-2.6	0.70
	Qwen3-Omni	0.65
CMM(Vis Dom)	Qwen2.5-Omni-7B	0.73
	MiniCPM-o-2.6	0.62
	Qwen3-Omni	0.57
CMM(Aud Dom)	Qwen2.5-Omni-7B	0.86
	MiniCPM-o-2.6	0.75
	Qwen3-Omni	0.87
AVHBench(V→A)	Qwen2.5-Omni-7B	0.57
	MiniCPM-o-2.6	0.57
	Qwen3-Omni	0.52
AVHBench(A→V)	Qwen2.5-Omni-7B	0.63
	MiniCPM-o-2.6	0.63
	Qwen3-Omni	0.72

Table 4 AUROC results across tasks and models. Qwen3-Omni refers to Qwen3-Omni-30B-A3B-Instruct.

$\alpha = -0.7$, the accuracy improves from the baseline of 72.14% to 81.53%. In contrast, positive coefficients consistently reduce accuracy. On Audio-driven Video Hallucination, suppressing representations along the distractor modality direction improves accuracy. The best performance is achieved at $\alpha = -0.3$. Stronger suppression decreases accuracy because the baseline preference for the audio modality is already low, such that further suppression yields no additional positive gains. Conversely, amplifying this direction leads to a decline, which drops to 80.90% at $\alpha = 0.7$. To further examine whether these effects result from nonspecific hidden-state perturbations, we evaluate the intervened model on OmniBench (Li et al., 2026), a general-purpose omni-modal understanding benchmark. For each steering direction, we directly adopt the α that achieves the best performance on the tasks and apply it at inference time, without additional tuning. The results are shown in Table 3. Suppressing the audio direction ($\alpha = -0.3$) leaves accuracy nearly unchanged (0.26%). Notably, suppressing the visual direction ($\alpha = -0.7$) increases accuracy by 6.67%. These results indicate that steering does not introduce nonspecific perturbations but instead yields beneficial effects.

6.4 Zero-shot Probe-based Hallucination Detection

We employ the linear probe to perform zero-shot hallucination detection across all 5 tasks. For each sample, we use the probe-estimated distractor modality preference probability as the hallucination risk score. We report AUROC, a threshold-independent measure of discrimination between hallucination and non-hallucination samples. The results are shown in Table 4.

Our probes enable effective zero-shot hallucination detection without requiring any downstream task data. Across all 5 tasks and 3 OLLMs with different architectures and scales, the probe-based method substantially outperforms the random baseline of 0.5 in most settings. In particular, Qwen2.5-Omni-7B achieves an AUROC of 0.86 on Audio Dominance, while Qwen3-Omni-30B-A3B-Instruct reaches 0.87, with an average AUROC of 0.83. These results demonstrate that the probes can transfer to hallucination detection in a zero-shot manner, indicating that the captured preference signals exhibit a certain degree of generalizability. Tasks with audio as the distractor modality achieve better detection performance. We attribute this result to the lower baseline audio preference, which makes hallucination-induced preference changes more pronounced. In contrast, the higher baseline visual preference increases the difficulty of tasks with vision as the distractor modality.

Figure 7 presents representative cases from 3 cross-modal hallucination tasks, each involving a different distractor modality: the Language Dominance subset of CMM (text as distractor), the Video-driven Audio

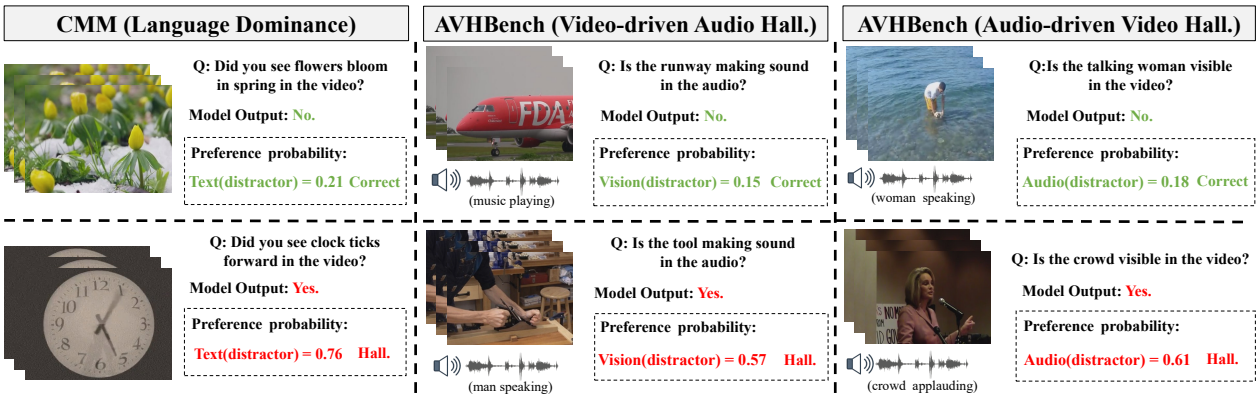


Figure 7 Case studies on 3 cross-modal hallucination tasks with different distractor modalities.

Hallucination task of AVHBench (vision as distractor), and the Audio-driven Video Hallucination task of AVHBench (audio as distractor). In the Language Dominance task, given a video showing snow-covered flowers, the model correctly answers "No" when asked whether the flowers bloom in spring, with a probe-estimated distractor-modality preference probability of only 0.21. In contrast, when asked whether the clock ticks in the video is moving forward, the model produces a cross-modal hallucination, accompanied by a substantial increase in the distractor-modality preference probability to 0.76. In the Video-driven Audio Hallucination task, the distractor-modality preference probability was only 0.15 when the model did not hallucinate, but rose to 0.57 when hallucination occurred. Similarly, in the Audio-driven Video Hallucination task, the distractor-modality preference probability was only 0.18 when the model answered correctly, but increased to 0.61 when the model hallucinated. Overall, hallucination samples exhibit substantially higher distractor modality preference probabilities than non-hallucination samples. These cases demonstrate that our probe effectively captures abnormal increases in the model's preference for the distractor modality and can serve as a practical zero-shot tool for cross-modal hallucination detection without requiring task-specific data.

7 Conclusion

In this paper, we systematically investigate modality preference in OLLMs. Our framework reveals that most OLLMs exhibit a pronounced visual preference. Layer-wise probing further reveals that modality preference progressively emerges in the mid-to-late layers. Finally, we demonstrate that modality preference causally contributes to cross-modal hallucinations and the preference signals can be effectively leveraged for hallucination detection. This work provides a novel perspective on the development of more trustworthy omni-modal artificial intelligence systems, and can inspire further research.

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